

RealDiffusion: Physics-informed Attention for Multi-character Storybook Generation

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Prompt 1: Art style of a classic comic book, two heroes, sharing pizza, racing to tower, watching city, celebrating.



Prompt 2: Ghibli style anime, brother and sister, running in grass, sharing watermelon, wading in stream, looking from train.



Prompt 3: Modern Chinese animation style, sitting, playing a melody, releasing floating lanterns, reading bamboo.



Figure 1. Our framework tunes the balance between storytelling dynamism (left column) and robust character coherence (right column).

Abstract

While modern diffusion models excel at generating diverse single images, extending this to sequential generation reveals a fundamental challenge: balancing narrative dynamism with multi-character coherence. Existing methods often falter at this trade-off, leading to artifacts where characters lose their identity or the story stagnates. To resolve this critical tension, we introduce RealDiffusion, a unified framework designed to reconcile robust coherence with narrative dynamism. Heat diffusion serves as a dissipative prior that averages neighboring features along the sequence and removes high-frequency noise within the subject region. This suppresses attribute drift and sta-

bilizes identity across frames. A region-aware stochastic process then introduces small perturbations that explore nearby modes and prevent collapse so the story maintains pose change and scene evolution. We thus introduce a lightweight, training-free Physics-informed Attention mechanism that injects controllable physical priors into the self-attention layers during inference. By modeling feature evolution as a configurable physical system, our method regularizes spatio-temporal relationships without suppressing intentional, prompt-driven changes. Extensive experiments demonstrate that RealDiffusion achieves substantial gains in character coherence while preserving narrative dynamism, outperforming state-of-the-art approaches. Code is available at <https://github.com/ShmilyQi-CN/RealDiffusion>.

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1. Introduction

In recent years, diffusion models have rapidly evolved, demonstrating remarkable potential in content generation tasks including image synthesis [4, 24, 28], 3D object generation [16, 21], and video production [1, 11, 29]. Leveraging large-scale pre-training on massive datasets, these models now outperform earlier approaches in producing high-fidelity and coherent generative content.

Current approaches range from slow fine-tuning methods like DreamBooth [26] and Textual Inversion [6], to zero-shot ID injection with encoders like IP-Adapter [38], PhotoMaker [15], and InstantID [36], but these sacrifice full-body coherence for facial fidelity. The most relevant line of work involves training-free methods that enforce coherence by manipulating internal features. This is achieved through shared self-attention [33, 40], addressing attention biases [35], prompt engineering [17], or novel sampling techniques [14]. Despite these advances, such methods frequently fail in intricate multi-character scenarios, resulting in artifacts like attribute swapping and illogical spatial layouts [7]. Our work distinguishes itself by specifically targeting these challenges in multi-character story generation.

Diverse image generation is largely a solved problem. The critical frontier is maintaining coherence, especially in multi-character scenarios with their inherent trade-off between narrative dynamism and character coherence. We draw inspiration from physical systems to resolve this. Heat diffusion offers a natural smoothing kernel to enforce coherence, while a stochastic process provides a complementary kernel to preserve dynamism. Unlike established physics-informed methods like PINNs that require costly training [23], our framework uses these principles as direct, training-free computational tools at inference.

We introduce RealDiffusion, a novel framework for mastering this trade-off, illustrated in Figure 2. It features a training-free Physics-informed Attention mechanism guided by dynamic masks. Its core is our PhysicsOperator, which employs an insulated heat diffusion kernel and a stochastic kernel in tandem. A **Global Trade-off Controller** α governs the system by strengthening heat diffusion and foundational ID injection, empowering users to select their desired balance between coherence and dynamism.

Our contributions are summarized as follows:

- **RealDiffusion:** a novel, training-free controllable framework that explicitly balances coherence–dynamism and excels on multi-character stories.
- **Physics-informed Attention:** a training-free, mask-guided self-attention mechanism; *insulated heat diffusion* smooths features and suppresses attribute drift, while *region-aware stochasticity* preserves evolution.
- **Results & metrics:** state-of-the-art storybook quality and new sequence-level metrics for coherence and dynamism.

2. Related work

2.1. Diffusion Models

Diffusion models [30] have become the dominant paradigm in generative modeling. Their formulation as Denoising Diffusion Probabilistic Models (DDPMs) [10] and their unification with score-based models via stochastic differential equations (SDEs) [32] provided a solid theoretical foundation. The introduction of Latent Diffusion Models (LDMs) [24] was a pivotal efficiency breakthrough, enabling high-resolution synthesis by operating in a compressed latent space. Architecturally, while early models relied on U-Net [25], recent state-of-the-art models increasingly adopt Transformer-based backbones [20], showcasing superior scaling properties. Controllability is a key strength, achieved through techniques like Classifier-Free Guidance [9] for better text alignment and external frameworks like ControlNet [39] for explicit spatial guidance.

2.2. Storybook Generation

Extending single-image diffusion to multi-frame storybooks foregrounds the challenge of temporal coherence, especially in multi-character settings. Prior work spans three main directions. (i) *Personalization-based* methods embed subject identity via test-time or few-shot adaptation—DreamBooth [26], Textual Inversion [6], and more efficient LoRA variants [12, 27] or multi-concept composition [13]—but require per-subject training and risk drift/overfitting. (ii) *Encoder-based ID injection* conditions generation on image-encoder features, with IP-Adapter [38] and successors such as PhotoMaker [15] and InstantID [36] improving identity fidelity yet sometimes sacrificing prompt faithfulness or full-body coherence. (iii) *Training-free* approaches intervene at inference: attention manipulation (StoryDiffusion [40], ConsiStory [33], CharaConsist [35]) and multi-subject composition (FastComposer [37]); prompt engineering that concatenates all frame descriptions with token reweighting (OnePromptOneStory [17]); and novel samplers such as Zigzag Sampling [14]. Despite progress, complex scenes still exhibit artifacts (e.g., attribute swapping or illogical layouts [7]) and most training-free methods lack an explicit, user-controllable knob to balance narrative dynamism and character coherence—a gap our RealDiffusion addresses.

2.3. Physics-informed Neural Networks (PINNs)

Physics-informed Neural Networks (PINNs) represent a class of deep learning models designed to solve partial differential equations (PDEs) [23]. They achieve this by embedding the physical laws described by the PDEs directly into the neural network’s loss function. This ensures the learned solution adheres to the specified physical dynamics, boundary conditions, and initial states. The stan-

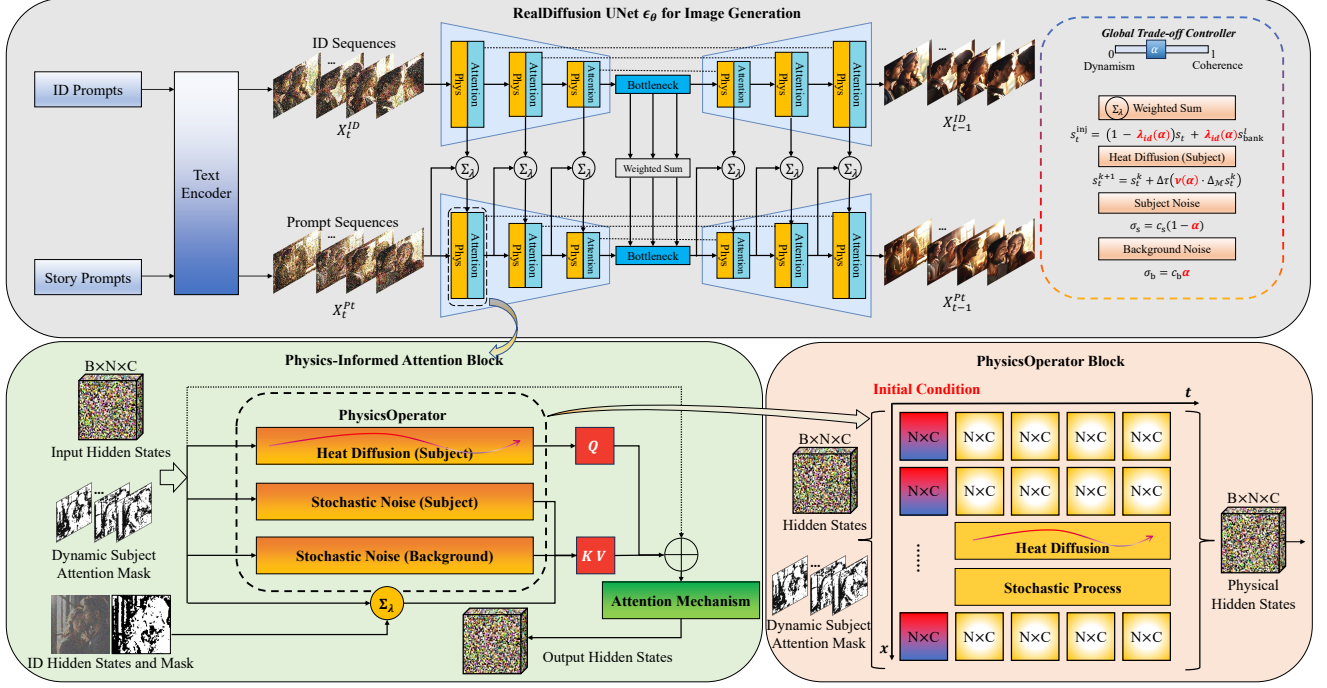


Figure 2. Overview of RealDiffusion. Our framework incorporates Physics-informed Attention into a U-Net, governed by α . The module combines insulated heat diffusion for coherence with region-aware stochasticity for dynamism, guided by dynamic subject masks.



Figure 3. An 8-frame story sequence with its dynamic masks. Prompt: A pencil sketch of, bride and groom; in morning fog, looking at horizon, on a stone bridge, walking side-by-side, under aurora, sharing a quiet moment, under lantern light, standing back-to-back.

dard PINN paradigm, however, requires retraining for any change in PDE parameters, a limitation discussed in recent surveys [2, 34]. To overcome this, operator learning and meta-learning approaches have been proposed to learn a mapping from PDE parameters to the solution function, allowing for near-instantaneous inference after an initial, costly training phase [3, 22]. Our work draws conceptual inspiration from this field, not by training a network to solve a PDE, but by using the mathematical principles of physical processes (specifically, heat diffusion) as a computational tool to regularize the generative process at inference time.

3. RealDiffusion

Rather than learning coherence, we control it at inference time. We view multi-frame generation as a controllable dynamical system and inject lightweight physics-based priors directly into attention. The overall architecture is shown in Figure 2. We introduce the essential notation here. The system operates on a sequence of per-frame hidden features $S = \{s_t\}_{t=1}^T$, where the subscript t denotes the frame index. These features are guided by a dynamic binary subject mask $\mathcal{M}_t$. The entire process is governed by $\alpha \in [0, 1]$. This knob simultaneously modulates four key parameters, each scaled by a base constant c . The full sampling proce-

Table 1. Physical priors for temporal regularization. The frame index t is treated as a spatial dimension, regularized over a virtual time τ .

Physical Law	Conceptual Basis / PDE Form	Discrete Update Rule (Variational Form)
Ori (Baseline)	N/A (No physical prior)	$s_t^{k+1} = s_t^k$
Burgers' Equation	$\frac{\partial s}{\partial \tau} + s \frac{\partial s}{\partial t} = 0$	$s_t^{k+1} = s_t^k - \Delta\tau \cdot s_t^k \frac{s_{t+1}^k - s_{t-1}^k}{2}$
Wave Equation	$\frac{\partial^2 s}{\partial \tau^2} = c^2 \frac{\partial^2 s}{\partial t^2}$	$s_t^{k+1} = 2s_t^k - s_t^{k-1} + C(s_{t+1}^k - 2s_t^k + s_{t-1}^k)$
Conservation Law	$\frac{\partial s}{\partial \tau} + \nabla_t \cdot \mathbf{F}(s) = 0$	$s_t^{k+1} = s_t^k - \Delta\tau \frac{F(s_{t+1}^k) - F(s_{t-1}^k)}{2}$
Elasticity	$\rho \frac{\partial^2 s}{\partial \tau^2} = \nabla_t \cdot \sigma$	$s_t^{k+1} = 2s_t^k - s_t^{k-1} + C_e(s_{t+1}^k - 2s_t^k + s_{t-1}^k)$
Ours (Heat Diff.)	$\frac{\partial s}{\partial \tau} = \nu(\alpha) \nabla_t^2 s$	$s_t^{k+1} = s_t^k + \Delta\tau \cdot \nu(\alpha)(s_{t+1}^k - 2s_t^k + s_{t-1}^k)$

ture is detailed in Algorithm 1.

3.1. Dynamic Mask Generation

To localize interventions to subjects, we produce a binary mask $\mathcal{M}_t$ at each sampling step without training. We capture cross-attention maps that correspond to the subject tokens, average them over a recent window to stabilize noise, and binarize the result using Otsu's method [19]. The mask sequence gates all subsequent operations, ensuring that regularization acts on the subject and avoids background leakage. Example masks are illustrated in Figure 3.

3.2. PhysicsOperator

The PhysicsOperator regularizes the temporal sequence $\{s_t\}$ via a dissipative coherence kernel and a region-aware dynamism kernel, tuned by α . The choice of the coherence kernel is critical. As detailed in Table 1, we select the Heat Diffusion equation because it is a dissipative system that guarantees a stable smoothing process towards equilibrium. This contrasts with conservative systems (e.g., Wave, Elasticity) that cause feature oscillations, and non-linear systems (e.g., Burgers') that risk unpredictable artifacts. This theoretical stability is reflected empirically, as heat diffusion offers a significantly wider and more robust parameter tuning range than the highly sensitive alternatives.

Coherence via insulated heat diffusion. To suppress high-frequency temporal incoherence, we use the heat equation as a dissipative prior, whose continuous form is:

$$\frac{\partial s(x, \tau)}{\partial \tau} = \nabla \cdot (\nu(\alpha) \nabla s(x, \tau)), \quad (1)$$

where x is the spatial position and τ is a virtual time. For the discrete feature sequence, we employ periodic boundary conditions, allowing the first and last frames to influence each other for temporal coherence. The core of our method is the *insulated* Laplacian $\Delta_{\mathcal{M}}$, which confines the smoothing process to the subject regions:

$$\Delta_{\mathcal{M}} s_t = s'_{t+1} - 2s_t + s'_{t-1}. \quad (2)$$

The insulation is achieved by defining the effective neighbors $s'_{t\pm 1}$ conditionally based on the dynamic masks:

$$s'_{t\pm 1} = \mathcal{M}_{t\pm 1} \odot s_{t\pm 1} + (1 - \mathcal{M}_{t\pm 1}) \odot s_t. \quad (3)$$

As defined in Eq. 3, if a neighboring feature at frame $t \pm 1$ is outside the subject mask (i.e., $\mathcal{M}_{t\pm 1}$ is zero), its value is replaced by the current feature s_t . This effectively creates a zero-flux boundary at the mask's edge, ensuring the diffusion process only occurs within the subject region and preventing information leakage from the background.

Dynamism via region-aware stochasticity. To avoid over-smoothing and preserve narrative dynamism, we inject controlled noise with region-aware intensities. The intensities differ inside (σ_s) and outside (σ_b) the mask $\mathcal{M}_t$ and are both governed by α . The subject noise intensity σ_s is inversely tied to α to stabilize the subject, while the background noise intensity σ_b is modulated by α to preserve scene dynamism. We combine these into a single, spatially-varying noise intensity tensor σ_t :

$$\sigma_t = \mathcal{M}_t \cdot \sigma_s + (1 - \mathcal{M}_t) \cdot \sigma_b. \quad (4)$$

This composite intensity σ_t is then used in the stochastic part of the full update rule (Eq. 5), providing a compact representation of our region-aware dynamism kernel.

Discrete update. Combining both kernels with a forward Euler step in τ , the per-iteration update is

$$s_t^{k+1} = s_t^k + \underbrace{\Delta\tau (\nu(\alpha) \cdot \Delta_{\mathcal{M}} s_t^k)}_{\text{coherence}} + \underbrace{\sqrt{2\Delta\tau} \cdot \sigma_t \mathcal{N}(0, I)}_{\text{dynamism}}, \quad (5)$$

where $\Delta\tau$ is the virtual time step and $\mathcal{N}(0, I)$ is standard normal noise. The coefficients introduced in Eq. 5 are simple, intuitive functions of the control parameter α . Their precise formulations are detailed in Algorithm 1.

Algorithm 1 RealDiffusion Sampling

Require: Prompt p ; frames T ; steps L ; control $\alpha \in [0, 1]$; ID bank $\{s_{\text{bank}}^l\}_{l=1}^L$; PhysicsOperator iters n ; virtual time step $\Delta\tau$.

1: **Notation:**

Q, K, V – Queries, Keys, and Values in attention.

s_t^{phys} – Physically-smoothed features.

s_t^{id} – Identity-enhanced features.

Ensure: Images $\{x_t^*\}_{t=1}^T$.

- 2: Encode p .
 - 3: For $t = 1..T$: sample $z_L^{(t)} \sim \mathcal{N}(0, I)$.
 - 4: Set $\nu \leftarrow c_{\text{heat}}\alpha$; $\lambda_{\text{id}} \leftarrow c_{\text{id}}\alpha$; $\sigma_s \leftarrow c_s(1 - \alpha)$; $\sigma_b \leftarrow c_b\alpha$.
 - 5: **for** $\ell = L, \dots, 1$ **do**
 - 6: Collect subject cross-attention maps.
 - 7: For each t : $\mathcal{M}_t \leftarrow \text{Otsu}(\text{AvgRecentAttn})$.
 - 8: Extract current hidden states $\{s_t\}_{t=1}^T$.
 - 9: Use σ_s inside $\mathcal{M}_t$, σ_b outside $\mathcal{M}_t$ to get σ_t by Eq. 4.
 - 10: Apply Eq. 5 n iters with step $\Delta\tau$ to get $\{s_t^{\text{phys}}\}$.
 - 11: $s_t^{\text{inj}} \leftarrow (1 - \lambda_{\text{id}})s_t + \lambda_{\text{id}}s_{\text{bank}}^l$; $s_t^{\text{id}} \leftarrow \mathcal{M}_t \odot s_t^{\text{inj}} + (1 - \mathcal{M}_t) \odot s_t$.
 - 12: Build Q from s_t^{phys} ; build (K, V) from s_t^{id} ; U-Net to get $\hat{e}_\ell$.
 - 13: Scheduler update: $z_{\ell-1}^{(t)}$ from $z_\ell^{(t)}$ and $\hat{e}_\ell$.
 - 14: **end for**
 - 15: Decode each $z_0^{(t)}$ with VAE to obtain x_t^* .
 - 16: **return** $\{x_t^*\}_{t=1}^T$.
-

Physics-informed attention. We integrate this process into attention via a disentangled intervention: Q are built from the PhysicsOperator-smoothed states to ensure temporal coherence, while (K, V) are built from identity-enhanced states that use an ID bank injection with a strength proportional to α . This allows attention to retrieve identity-faithful information using temporally smooth queries. The precise assignments are detailed in Algorithm 1.

4. Experiments

This section experimentally validates RealDiffusion’s effectiveness and controllability. We compare our method against state-of-the-art approaches, conduct ablation studies on the physical prior and the α controller, and introduce novel metrics to quantitatively measure the balance between coherence and dynamism.

4.1. Experimental Setup

Implementation Details. All experiments are built upon Stable Diffusion XL and run on a single NVIDIA RTX 3090. For fair comparison, all methods use a 1024×1024 resolution, 50 DDIM [31] sampling steps, and a 7.5 classifier-free guidance scale. For our PhysicsOperator, we use $n = 10$ iterations with a virtual time step of $\Delta\tau = 0.1$. The main experiments use $\alpha = 0.5$, and the base coefficients are set to $c_{\text{id}} = 1$, $c_{\text{heat}} = 2$, $c_s = 0.1$, and $c_b = 0.1$.

Metrics. **CLIP-T** measures text alignment using the average CLIPScore [8]. For identity preservation, **IDSIM** computes the average CLIP image similarity between each frame and the reference ID image. For temporal smoothness, **CLIP-I** measures adjacent-frame cosine similarity, while **DreamSim** [5] reports the perceptual distance.

To capture the trade-off central to our work, we introduce two sequence-aware measures on the sequence of per-frame CLIP features $\{f_t\}_{t=1}^T$: **Temporal Regularity** ($\mathcal{R}_t$) and **Storytelling Quality** ($\mathcal{S}_t$). $\mathcal{R}_t$ (lower is better) is designed to measure **coherence** by quantifying the smoothness of the feature trajectory. It penalizes temporal fluctuations by calculating the average L2 norm of the second-order differences, which acts as a discrete Laplacian:

$$\mathcal{R}_t = \frac{1}{T-2} \sum_{t=2}^{T-1} \|f_{t+1} - 2f_t + f_{t-1}\|_2. \quad (6)$$

Our second metric, $\mathcal{S}_t$, is designed to holistically evaluate the **balance between coherence and dynamism**. It is calculated by first defining an auxiliary first-order variation, $\mathcal{D}_t = \frac{1}{T-1} \sum_{t=1}^{T-1} \|f_{t+1} - f_t\|_2$, to quantify dynamism. We then convert $(\mathcal{R}_t, \mathcal{D}_t)$ into bounded scores $\hat{\mathcal{R}}_t = \exp(-\gamma_r \mathcal{R}_t)$ and $\hat{\mathcal{D}}_t = 1 - \exp(-\gamma_d \mathcal{D}_t)$, using small positive scaling constants γ_r and γ_d . $\mathcal{S}_t$ (higher is better) then uses a soft-min aggregator. This function approaches the minimum of the two scores at high p values and acts as a balanced mean for moderate values. We set $p = 4$, $\gamma_r = \gamma_d = 0.1$ in all experiments:

$$\mathcal{S}_t^{(p)} = \left(\frac{\hat{\mathcal{R}}_t^{-p} + \hat{\mathcal{D}}_t^{-p}}{2} \right)^{-1/p}. \quad (7)$$

4.2. Qualitative Results

Comparison with State-of-the-Art. Our evaluation, conducted on a comprehensive test set of 300 multi-character prompts generated by GPT-5 [18], highlights RealDiffusion’s strengths in complex scenarios. The qualitative results in Figure 4 show that our method generates sequences with significantly higher character coherence. While baseline methods often suffer from attribute swapping or identity drift in multi-character interactions, our approach maintains robust visual fidelity, preserving individual character features across diverse scenes. This focus on coherence is validated by our quantitative analysis in Table 2. RealDiffusion achieves the best performance on our proposed Storytelling Quality ($\mathcal{S}_t$) metric, which specifically rewards a balance between coherence and dynamism. This indicates that our method excels at managing this crucial trade-off. These gains in narrative coherence are achieved with minimal added computational cost in time and VRAM usage, confirming the lightweight nature of our framework.

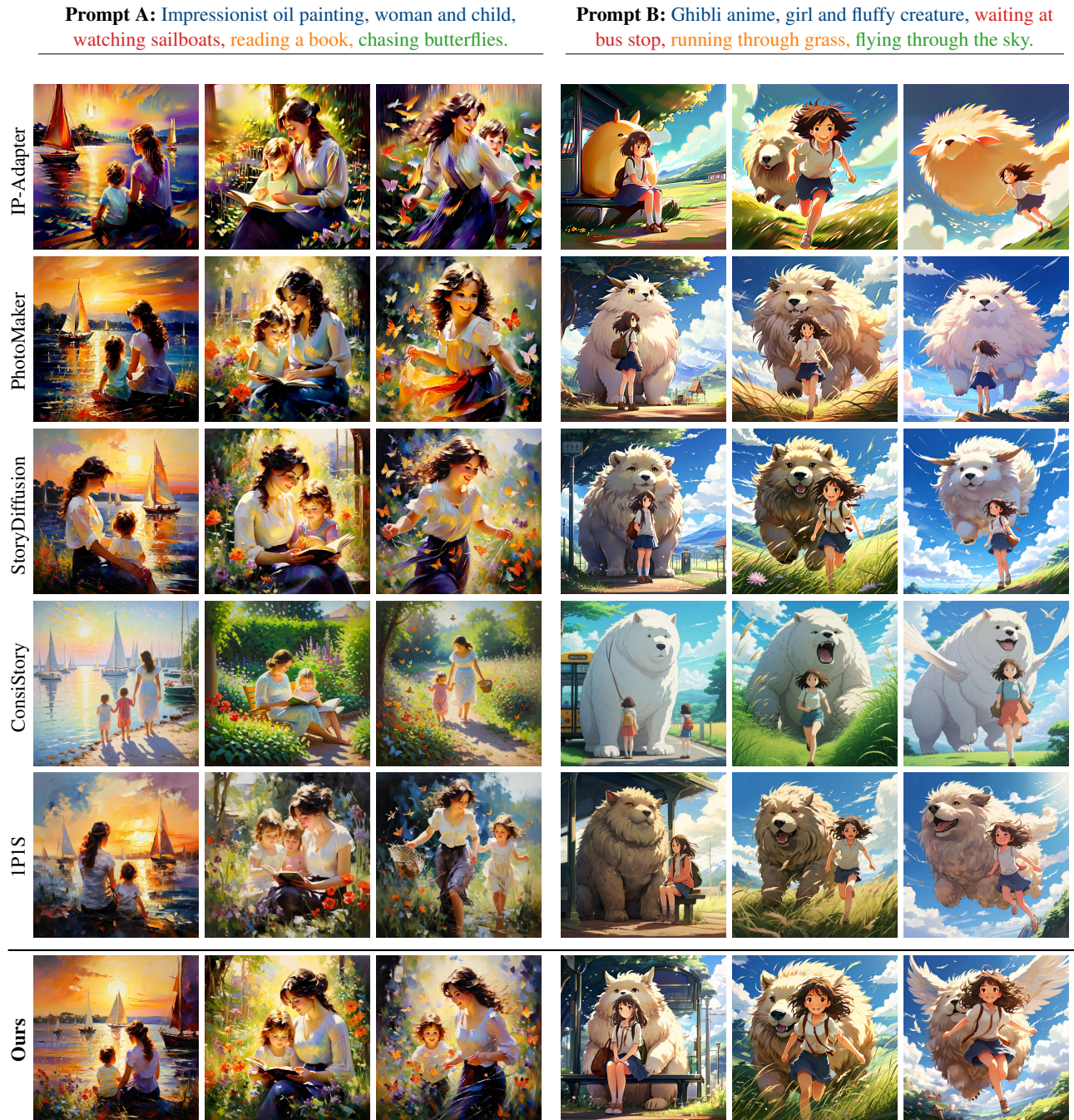


Figure 4. A qualitative comparison of our RealDiffusion against five state-of-the-art baseline models.

4.3. Ablation Studies

Choice of Physical Prior. An ablation study, run with $\alpha = 0.5$, validates our choice of the heat diffusion equation. As shown in Figure 5 and Table 3, heat diffusion is the most effective prior. Conservative systems like the Wave equation cause unstable feature oscillations, while non-linear systems like Burgers’ risk unpredictable arti-

facts. In contrast, the dissipative nature of heat diffusion provides a smooth and stable regularization, enhancing coherence without visual degradation.

Impact of the Global Trade-off Controller α . A core feature of our framework is the intuitive and powerful controllability offered by α . As illustrated in Figure 6, users can intuitively tune the balance between narrative dynamism and

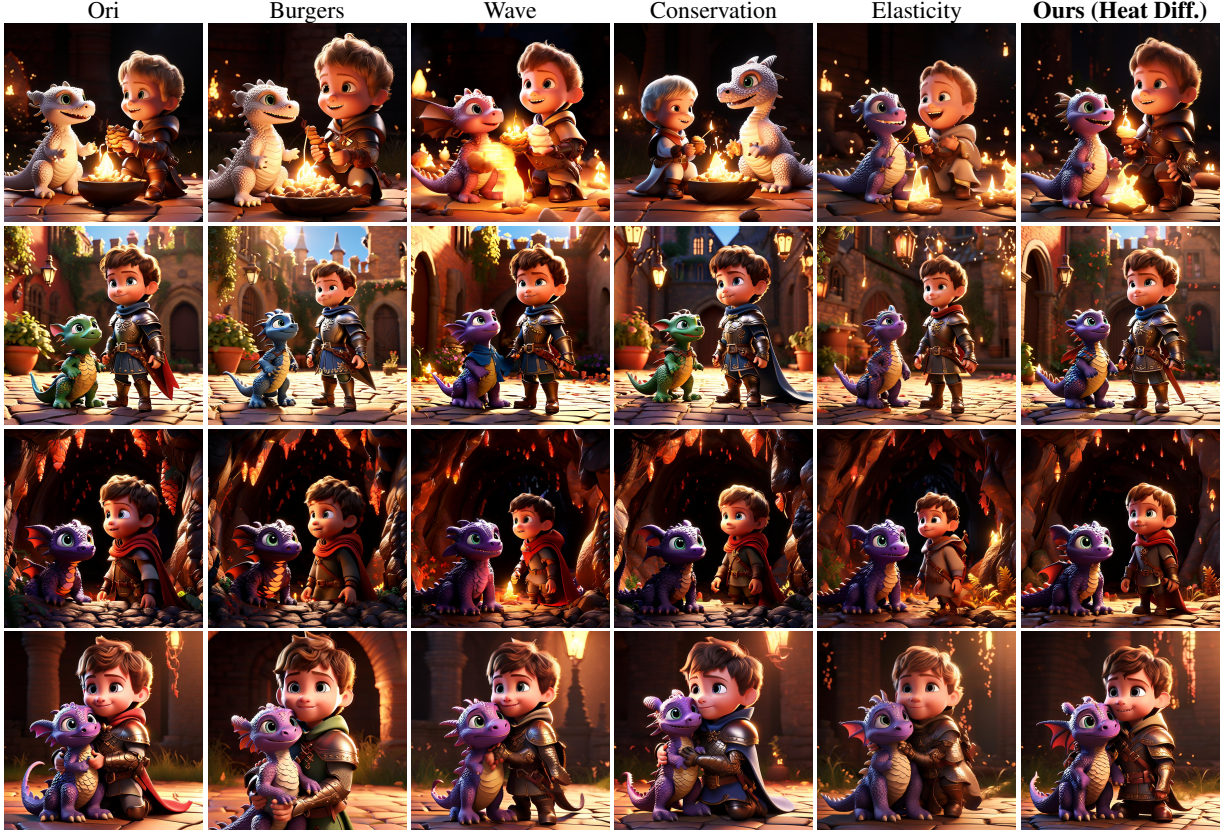


Figure 5. Ablation on physical priors. A high level of character coherence can be readily achieved by using heat diffusion as the coherence kernel. Prompt: **A 3d art style of Pixar, knight and baby dragon; roasting marshmallow, practicing, peeking into cave, hugging each other.**

Table 2. Quantitative comparison with baselines. ‘↑’ higher is better, ‘↓’ lower is better. **Bold**: best.

Method	Standard Metrics			Proposed Metrics		Cost Metrics	
	CLIP-T ↑	CLIP-I ↑	DreamSim ↓	$\mathcal{R}_t$ ↓	$\mathcal{S}_t$ ↑	Time (s)	VRAM (GB)
IP-Adapter [38]	0.2950	0.7664	0.3768	21.2323	0.1527	14.0	18.4
PhotoMaker [15]	0.2770	0.7826	0.3536	20.2335	0.1691	14.1	17.2
StoryDiffusion [40]	0.3121	0.8011	0.3286	19.2786	0.1837	16.7	20.4
ConsiStory [33]	0.3355	0.7905	0.3560	19.7387	0.1742	29.6	17.9
OnePromptOneStory [17]	0.3103	0.7860	0.3823	20.2241	0.1920	20.8	18.4
Zigzag Sampling [14]	0.3172	0.7706	0.3898	20.7667	0.1621	15.2	16.4
CharaConsist [35]	0.3167	0.7715	0.3847	20.6927	0.1617	15.5	16.5
RealDiffusion (Ours)	0.3161	0.8242	0.2780	17.9104	0.2132	20.5	17.8

character coherence. A low α prioritizes dynamic poses and actions, while a high α strengthens the physical prior to enforce robust visual fidelity. This crucial flexibility is notably absent in previous approaches. This trade-off is quantitatively validated in Figure 7, which presents our normalized metrics. The plot demonstrates that α allows for adjusting the balance between dynamism and coherence; selecting a value near 0.5, for instance, allows $\mathcal{S}_t$ to reach a favorable level without significantly compromising CLIP-T.

4.4. Discussion

Our results demonstrate that RealDiffusion makes a deliberate trade-off. While accepting a minimal computational overhead and a slight decrease in CLIP-T, our method excels on metrics measuring temporal coherence in CLIP-I, DreamSim and our proposed $\mathcal{R}_t$. This confirms our framework effectively prioritizes the core challenge of multi-character coherence. The ablation studies underscore the necessity of our chosen physical prior. The dissipative na-



Figure 6. Effect of α on the trade-off between coherence and dynamism. Prompt: Storybook watercolor, girl and bear, flying a kite, sharing honey, sitting on the moon, catching stars, sliding down a rainbow, and napping on a cloud.

Table 3. Ablation study on the choice of physical prior. Heat diffusion achieves the best performance.

Physical Law	Standard Metrics				Proposed Metrics	
	CLIP-T	CLIP-I	IDSIm	DreamSim	$\mathcal{R}_t$	S_t
Ori (Baseline)	0.3195	0.7812	0.8133	0.3603	20.3531	0.1670
Burgers' Equation	0.3077	0.7753	0.8146	0.3550	20.8394	0.1581
Wave Equation	0.3092	0.7835	0.7913	0.3725	20.2250	0.1671
Conservation Law	0.2969	0.7574	0.7842	0.3554	20.2755	0.1666
Elasticity	0.3106	0.7788	0.7918	0.3826	21.6029	0.1527
Ours (Heat Diff.)	0.3161	0.8242	0.8564	0.2780	17.9104	0.2132

ture of the heat diffusion equation is essential for guiding the generation towards a stable, coherent state. In contrast, conservative systems induce feature oscillations and non-linear priors introduced instability, confirming that a stable, dissipative process is crucial for temporal regularization. Finally, the controllability of our framework is centered on the parameter α . Our experiments reveal its practical implications: a very high α can lead to static characters lacking narrative dynamism, while a very low α diminishes the prior's influence, causing a loss of coherence. This shows α is an intuitive tool for navigating the fundamental trade-off between dynamism and coherence.

5. Conclusion

We introduce RealDiffusion, a novel framework specifically designed to tackle the critical challenge of coherent multi-character storybook generation. Our core is a Physics-informed Attention mechanism that injects a dual physical

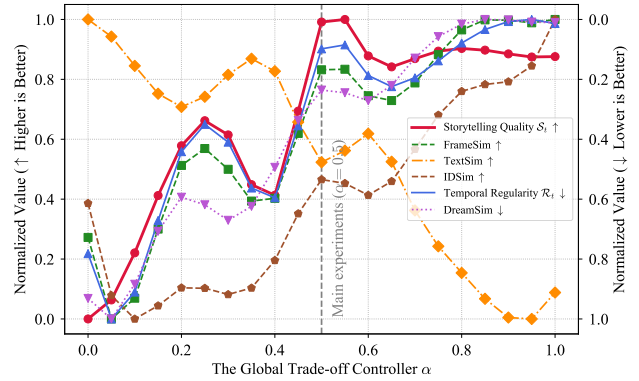


Figure 7. Quantitative impact of the controller α on metrics.

prior directly during inference. Heat diffusion acts as a dissipative prior suppressing attribute drift and identity swapping common in multi-character scenes. A complementary stochastic process preserves narrative dynamism allowing for natural character evolution and interaction. This dual system provides fine-grained control over the trade-off between multi-character coherence and narrative dynamism. The entire intervention operates at inference time avoiding costly retraining and ensuring seamless integration with standard, pre-trained SDXL backbones. Extensive experiments on complex multi-character prompts demonstrate substantial gains in identity coherence and sustained narrative dynamism across frames. Our work thus paves the way for controllable and coherent long-form visual storytelling.

Acknowledgment

This work was supported by the Key R&D Program of Zhejiang (No. 2026C02A1234). This work was completed during the internship of Qi Zhao at SGIT AI Lab, State Grid Corporation of China. The authors would like to thank the anonymous reviewers for their constructive comments.

References

- [1] Andreas Blattmann, Robin Rombach, Huan Ling, Tim Dockhorn, Seung Wook Kim, Sanja Fidler, and Karsten Kreis. Align your latents: High-resolution video synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 22563–22575, 2023. 2
- [2] Salvatore Cuomo, Vincenzo Schiano Di Cola, Fabio Giampaolo, Gianluigi Rozza, Maziar Raissi, and Francesco Piccialli. Scientific machine learning through physics-informed neural networks: Where we are and what’s next. *Journal of Scientific Computing*, 92(3):88, 2022. 3
- [3] Shaan Desai, Marios Mattheakis, Hayden Joy, Pavlos Protopapas, and Stephen Roberts. One-shot transfer learning of physics-informed neural networks. *arXiv preprint arXiv:2110.11286*, 2021. 3
- [4] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first international conference on machine learning*, 2024. 2
- [5] Stephanie Fu, Netanel Tamir, Shobhita Sundaram, Lucy Chai, Richard Zhang, Tali Dekel, and Phillip Isola. Dreamsim: Learning new dimensions of human visual similarity using synthetic data. *arXiv preprint arXiv:2306.09344*, 2023. 5
- [6] Rinon Gal, Yuval Alaluf, Yuval Atzmon, Or Patashnik, Amit H Bermano, Gal Chechik, and Daniel Cohen-Or. An image is worth one word: Personalizing text-to-image generation using textual inversion. *arXiv preprint arXiv:2208.01618*, 2022. 2
- [7] Yuan Gong, Youxin Pang, Xiaodong Cun, Menghan Xia, Yingqing He, Haoxin Chen, Longyue Wang, Yong Zhang, Xintao Wang, Ying Shan, et al. Talecrafter: Interactive story visualization with multiple characters. *arXiv preprint arXiv:2305.18247*, 2023. 2
- [8] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning. *arXiv preprint arXiv:2104.08718*, 2021. 5
- [9] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022. 2
- [10] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020. 2
- [11] Jonathan Ho, William Chan, Chitwan Saharia, Jay Whang, Ruiqi Gao, Alexey Gritsenko, Diederik P Kingma, Ben Poole, Mohammad Norouzi, David J Fleet, et al. Imagen video: High definition video generation with diffusion models. *arXiv preprint arXiv:2210.02303*, 2022. 2
- [12] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022. 2
- [13] Nupur Kumari, Bingliang Zhang, Richard Zhang, Eli Shechtman, and Jun-Yan Zhu. Multi-concept customization of text-to-image diffusion. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1931–1941, 2023. 2
- [14] Mingxiao Li, Mang Ning, and Marie-Francine Moens. Consistent story generation: Unlocking the potential of zigzag sampling. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. 2, 7
- [15] Zhen Li, Mingdeng Cao, Xintao Wang, Zhongang Qi, Ming-Ming Cheng, and Ying Shan. Photomaker: Customizing realistic human photos via stacked id embedding. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 8640–8650, 2024. 2, 7
- [16] Chen-Hsuan Lin, Jun Gao, Luming Tang, Towaki Takikawa, Xiaohui Zeng, Xun Huang, Karsten Kreis, Sanja Fidler, Ming-Yu Liu, and Tsung-Yi Lin. Magic3d: High-resolution text-to-3d content creation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 300–309, 2023. 2
- [17] Tao Liu, Kai Wang, Senmao Li, Joost van de Weijer, Fahad Shahbaz Khan, Shiqi Yang, Yaxing Wang, Jian Yang, and Ming-Ming Cheng. One-prompt-one-story: Free-lunch consistent text-to-image generation using a single prompt. *arXiv preprint arXiv:2501.13554*, 2025. 2, 7
- [18] OpenAI. Gpt-5 system card. <https://openai.com/index/gpt-5-system-card/>, 2025. Accessed 2025-11-12. 5
- [19] Nobuyuki Otsu. A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1):62–66, 1979. 4
- [20] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4195–4205, 2023. 2
- [21] Ben Poole, Ajay Jain, Jonathan T Barron, and Ben Mildenhall. Dreamfusion: Text-to-3d using 2d diffusion. *arXiv preprint arXiv:2209.14988*, 2022. 2
- [22] Apostolos F Psaros, Kenji Kawaguchi, and George Em Karniadakis. Meta-learning pinn loss functions. *Journal of computational physics*, 458:111121, 2022. 3
- [23] Maziar Raissi, Paris Perdikaris, and George E Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational physics*, 378:686–707, 2019. 2
- [24] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022. 2

- [25] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer, 2015. [2](#)
- [26] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 22500–22510, 2023. [2](#)
- [27] Simo Ryu. Low-rank adaptation for fast text-to-image diffusion fine-tuning. *Low-rank adaptation for fast text-to-image diffusion fine-tuning*, 3, 2023. [2](#)
- [28] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily L Denton, Kamyar Ghasemipour, Raphael Gontijo Lopes, Burcu Karagol Ayan, Tim Salimans, et al. Photorealistic text-to-image diffusion models with deep language understanding. *Advances in neural information processing systems*, 35:36479–36494, 2022. [2](#)
- [29] Uriel Singer, Adam Polyak, Thomas Hayes, Xi Yin, Jie An, Songyang Zhang, Qiyan Hu, Harry Yang, Oron Ashual, Oran Gafni, et al. Make-a-video: Text-to-video generation without text-video data. *arXiv preprint arXiv:2209.14792*, 2022. [2](#)
- [30] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *International conference on machine learning*, pages 2256–2265. pmlr, 2015. [2](#)
- [31] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020. [5](#)
- [32] Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. *arXiv preprint arXiv:2011.13456*, 2020. [2](#)
- [33] Yoad Tewel, Omri Kaduri, Rinon Gal, Yoni Kasten, Lior Wolf, Gal Chechik, and Yuval Atzmon. Training-free consistent text-to-image generation. *ACM Transactions on Graphics (TOG)*, 43(4):1–18, 2024. [2](#), [7](#)
- [34] Edgar Torres, Jonathan Schiefer, and Mathias Niepert. Adaptive physics-informed neural networks: A survey. *arXiv preprint arXiv:2503.18181*, 2025. [3](#)
- [35] Mengyu Wang, Henghui Ding, Jianing Peng, Yao Zhao, Yunpeng Chen, and Yunchao Wei. Characonsistent: Fine-grained consistent character generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16058–16067, 2025. [2](#), [7](#)
- [36] Qixun Wang, Xu Bai, Haofan Wang, Zekui Qin, Anthony Chen, Huaxia Li, Xu Tang, and Yao Hu. Instantid: Zero-shot identity-preserving generation in seconds. *arXiv preprint arXiv:2401.07519*, 2024. [2](#)
- [37] Guangxuan Xiao, Tianwei Yin, William T Freeman, Frédo Durand, and Song Han. Fastcomposer: Tuning-free multi-subject image generation with localized attention. *International Journal of Computer Vision*, 133(3):1175–1194, 2025. [2](#)
- [38] Hu Ye, Jun Zhang, Sibio Liu, Xiao Han, and Wei Yang. Ip-adapter: Text compatible image prompt adapter for text-to-image diffusion models. *arXiv preprint arXiv:2308.06721*, 2023. [2](#), [7](#)
- [39] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 3836–3847, 2023. [2](#)
- [40] Yupeng Zhou, Daquan Zhou, Ming-Ming Cheng, Jiashi Feng, and Qibin Hou. Storydiffusion: Consistent self-attention for long-range image and video generation. *Advances in Neural Information Processing Systems*, 37: 110315–110340, 2024. [2](#), [7](#)